

Computability & Logic, Exercises to Lectures 3+4

Aïna Linn Georges and Jaco van de Pol

Theoretical Exercises

From John C. Martin book

Advice

The material might be slightly too much for the exercise session, but with a proper preparation you should get quite far. Make sure you do at least some exercises in each category. The minimal sequence would be 3.18, 3.25, 3.37(e), 3.38(a), 3.40(b), 3.41(a), 3.49, 3.20(b), 3.51(b).

The other exercises are also relevant. Feel free to choose more interesting challenge problems from the book, when time permits!

NFA:

- 3.18
- 3.19
- 3.25
- 3.22. Draw the automaton. Use the definition of Λ and δ^* .

NFA to DFA:

- 3.37 (c, e)
- 3.38 (a,e)
- 3.40 (b) (you may do this at once, using the algorithm from the lecture slides)

RE to NFA:

- 3.41 (a,e)
- 3.49 (a,b)

NFA to RE:

- 3.20 (b) You may use Method 1 from Lecture 4 (slides), i.e. C. 6.2 from [Sudkamp]
- 3.51 (b) You may use Method 1 from Lecture 4 (slides), i.e. C. 6.2 from [Sudkamp]
- 3.54 (b) This is also Method 1 from Lecture 4 (slides), i.e. C. 6.2 from [Sudkamp]